TECHNIQUES OF DATA4LLM: Data Management for Large Language Models

Section 1.1 – A Survey of LLM × DATA

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MOTIVATION (Why DATA4LLM?)

Problem:

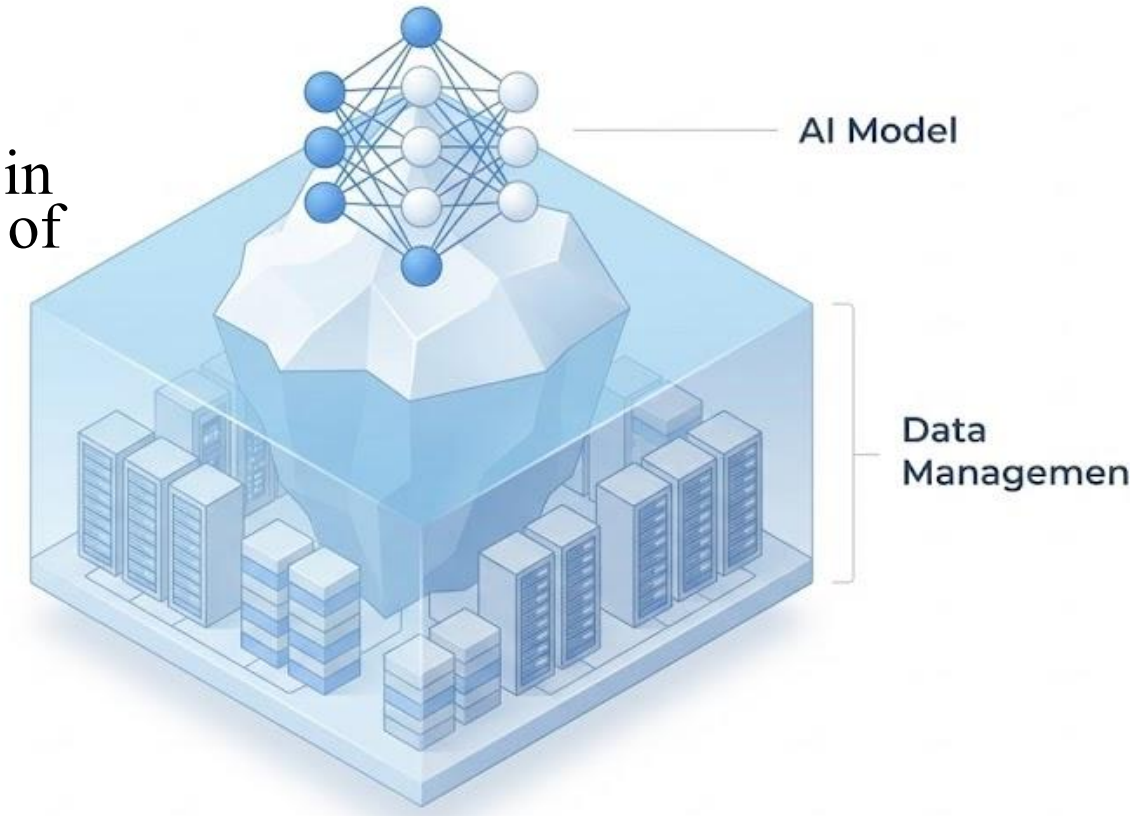
- Large Language Models (LLMs) are not just about architecture (Transformer, etc.) and parameters.
- Traditional data management approaches fall short in meeting the massive and heterogeneous data needs of LLMs .

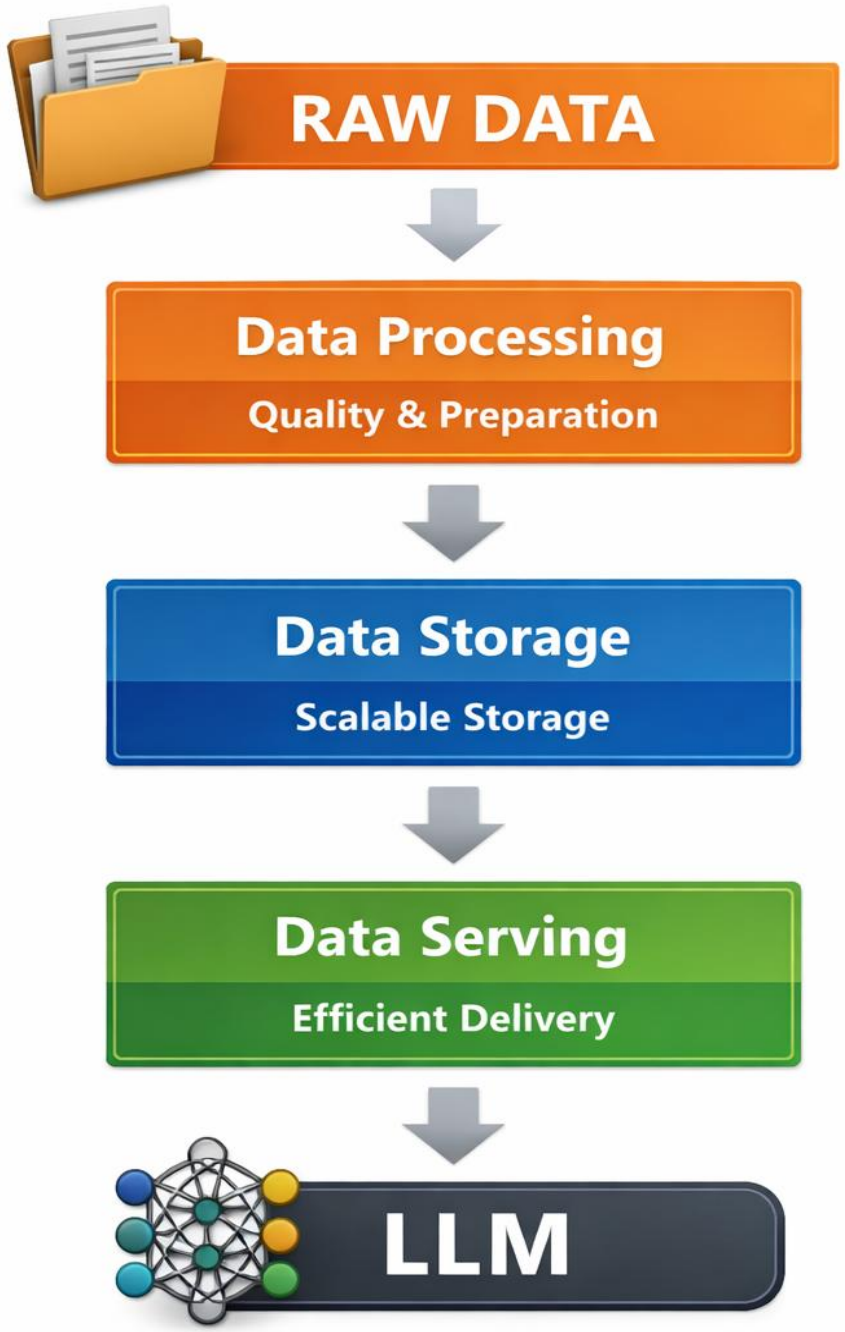
Core Idea:

- LLM performance depends directly on how data is **processed**, **stored**, and **served** to the model.

Equation:

- **High Performance = Strong Model + Effective DATA4LLM**




RAW DATA
Data Processing
Quality & Preparation
Data Storage
Scalable Storage
Data Serving
Efficient Delivery
LLM

WHAT IS DATA4LLM?

Definition:

- DATA4LLM is the collection of **all data management techniques** that feed and support the LLM lifecycle .

3 Key Components:

1. **Data Processing:** Making data ready for training (Cleaning, Mixing).
2. **Data Storage:** Storing data and models in a scalable manner.
3. **Data Serving:** Serving data during training and inference stages.

THE "IaaS" CONCEPT FOR DATA QUALITY

The paper defines LLM data quality across four fundamental dimensions (Not 'Infrastructure as a Service', but a data quality framework):

1. **Inclusiveness :** Covering diverse domains, languages, and modalities (text, image, code).
2. **Abundance :** Sufficient volume and balanced diversity for generalization capability .
3. **Articulation :** Well-formatted, clean, logical, and expressive data .
4. **Sanitization :** Data cleared of PII, toxic content, and biases .

📌 **Note:** These four dimensions form the foundation for evaluating DATA4LLM techniques.

DATA4LLM FRAMEWORK

COMPONENT 1: DATA PROCESSING

Goal: To transform raw data into an IaaS-compliant training set .

Critical Steps:

•**Acquisition & Deduplication :** Collecting data and cleaning duplicates to prevent the model from memorizing .

•**Filtering:** Eliminating low-quality, toxic content, and PII (using perplexity, content filters, etc.) .

•**Data Synthesis:** Generating synthetic data for data-scarce domains or to improve **reasoning** capabilities .

📌 **Note:** *Low-quality data inevitably leads to low-quality models.*

Processing Techniques: Algorithms & Methods

•**Deduplication :**

•**Exact Deduplication:** MD5/SHA-256 hash matching.

•**Near-Deduplication:** Uses **MinHash** & **LSH (Locality Sensitive Hashing)** to find semantic duplicates (Jaccard similarity).

•*Impact:* Reduces training cost, prevents memorization.

•**Data Selection Strategy :**

•**Quality Filtering:** Rules-based (heuristics) + Model-based (classifiers).

•**Diversity Mixing:** Balancing domain weights (e.g., 50% Web, 30% Code, 20% Books) to avoid domain collapse.

•**Synthetic Augmentation:**

•Using LLMs (Self-Instruct) to generate data for low-resource domains.

1
DEDUPLICATION
2
FILTERING & SELECTION
3
DOMAIN MIXING
TRAINING-READY CORPORA
www
www
web
code

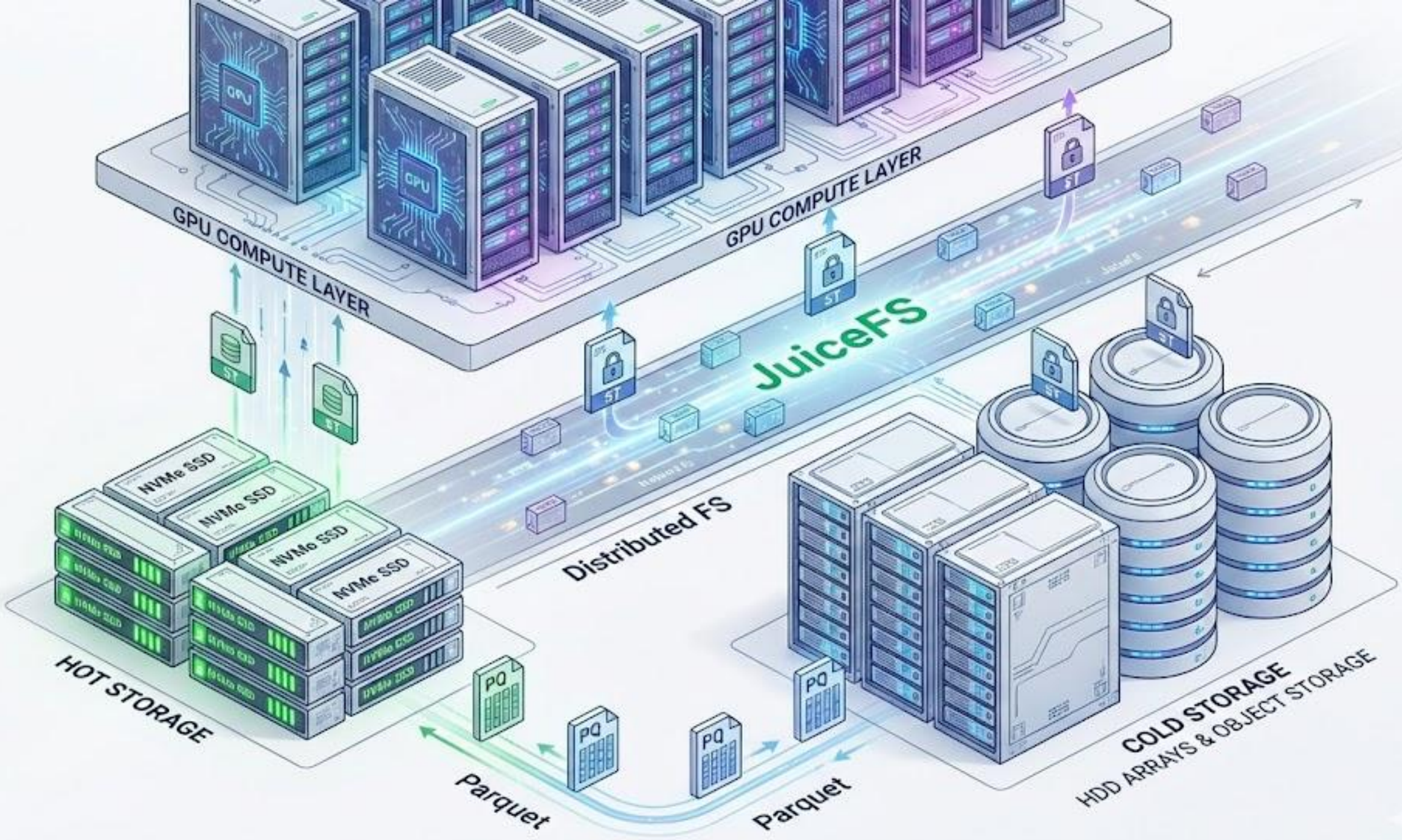
COMPONENT 2: DATA STORAGE

Problem:

- Training data at the TB–PB scale.
- Models with hundreds of billions of parameters.
- The need for high bandwidth between GPU–CPU–Disk.

Solutions:

- **Distributed Storage:** Managing data and models by splitting them across nodes (e.g., 3FS).
- **KV-Cache (**Key-Value Cache)**Management:** Optimizing inference memory, especially in **long-context** models.
- **Fault Tolerance:** Checkpointing mechanisms against training interruptions.


GPU COMPUTE LAYER
GPU COMPUTE LAYER
JuiceFS
Distributed FS
NVMe SSD
NVMe SSD
NVMe SSD
NVMe SSD
HOT STORAGE
COLD STORAGE
HDD ARRAYS & OBJECT STORAGE
Parquet
Parquet

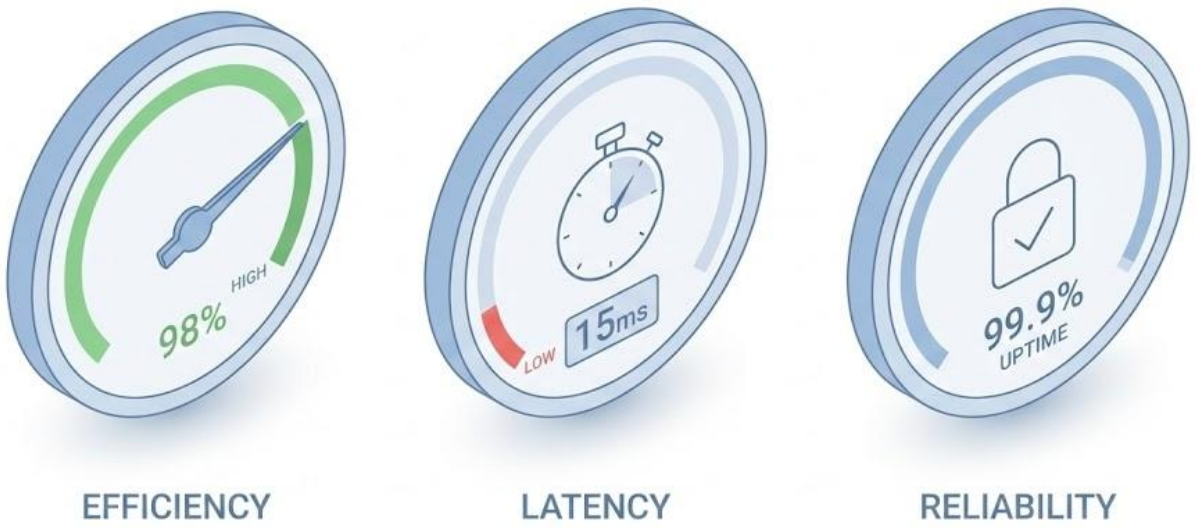
WHY IS DATA STORAGE CRITICAL?

- **Training Speed:** Directly depends on storage I/O throughput.
- **Inference Latency:** KV-cache management (latency) is the determining factor in long-context models.
- **Reliability:** Recovery after failure is vital in trainings that last for weeks.

Conclusion: Storage is no longer a passive repository, but an **active performance component.**

COMPONENT 3: DATA SERVING

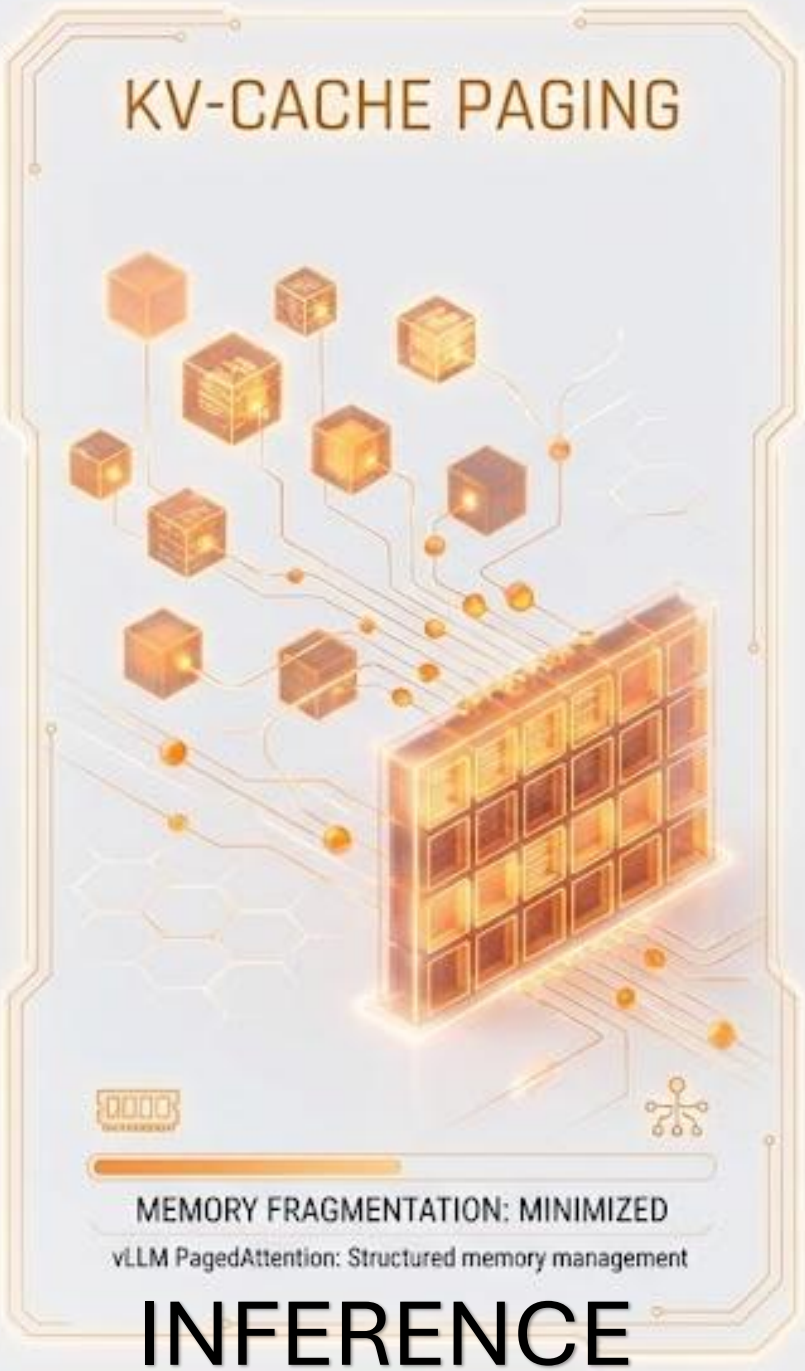
Goal: To present the **right data, at the right time, in the right format** to the model.



Critical Use Cases:

- **RAG (Retrieval-Augmented Generation) :** Integrating external information to prevent hallucinations.
- **Inference:** Efficient processing of prompts.
- **Training Efficiency:** Optimizing data order and ratios (Shuffling & Packing).

TRAINING



INFERENCE

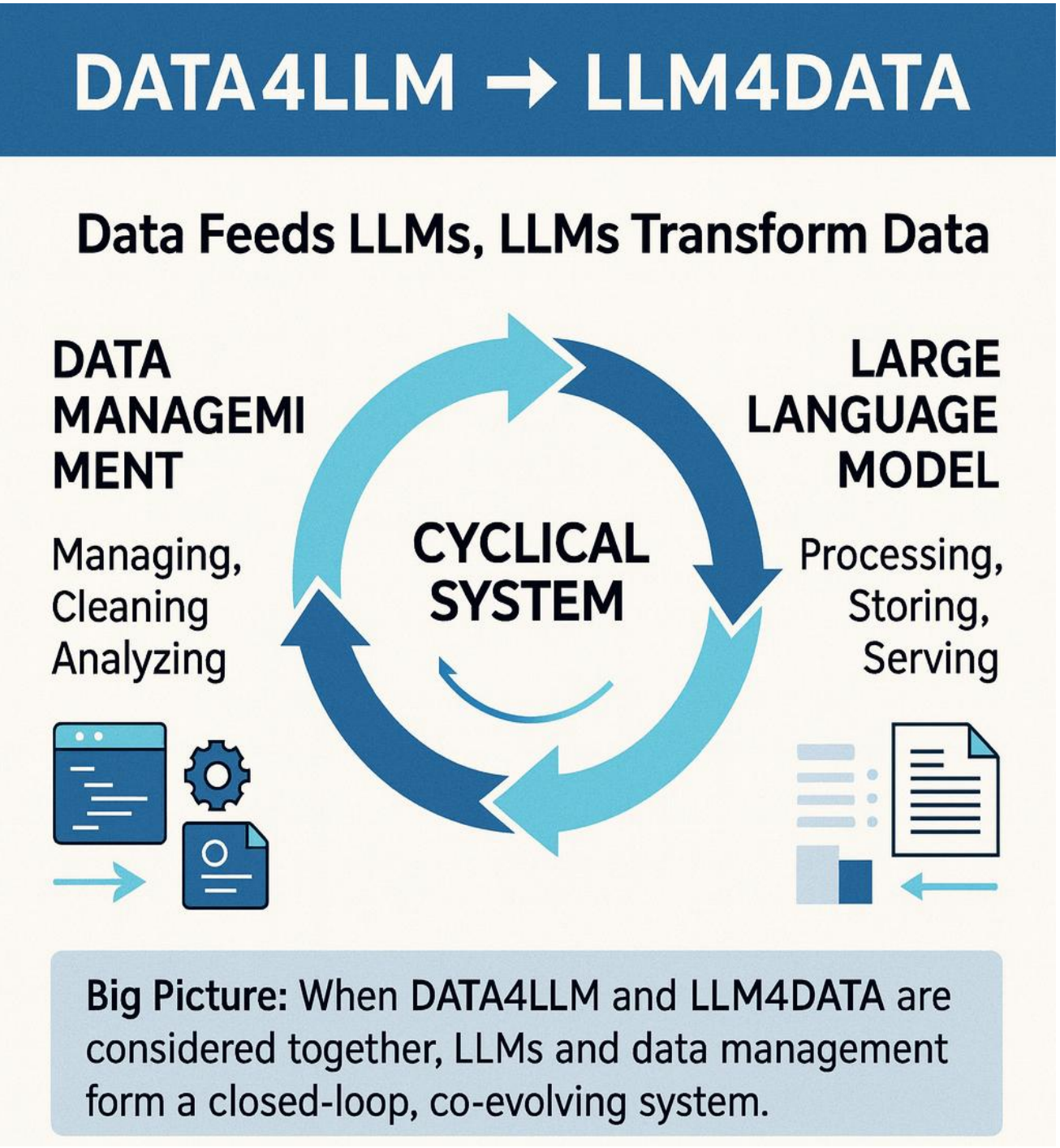
RAG

CONCLUSION AND SUMMARY

Section 1.1 Main Message:

- Developing an LLM is not just about building the model architecture.
- For a successful LLM, data must be:
  - Meticulously **processed**.
  - Stored in a **scalable** manner.
  - **Optimized** and served.

Final Word: DATA4LLM is the foundation of the reliability and performance of AI systems.



FUTURE VISION

- **DATA4LLM → LLM4DATA**
- **Data Feeds LLMs, LLMs Transform Data**
- This presentation focused on **DATA4LLM** *(Data → LLM)*.
- However, the relationship is **not one-directional**.

- **LLM4DATA (LLMs Managing Data)**
- Data Cleaning & Integration
- Data Analysis (Natural Language → SQL / Code)
- System Optimization